

Skyworth WiFi6 ONT Internet Hub AX3000S Specification

Revision History

Rev	Date	Revision History	Author
V1.0	30/9/2025	1 st version released	Qian Li
V2.0	9/10/2025	Update the pictures of Device Outlook, Hardware Diagram, LED indicators ,Bottom Panel and Rear Panel.	Qian Li
V3.0	18/11/2025	Update Key Components flash and DDR Vendor	Xu Hui
V4.0	20/11/2025	Update Housing (same appearance as AX3000S)	Qian Li
V5.0	26/11/2025	Update the pictures of Device Outlook. Add alternative power adapter supplier.	Qian Li
V6.0	13/01/2026	Update the Device Outlook and Housing.	Qian Li

CONTENT

1	PRODUCT INTRODUCTION	4
1.1	OVERVIEW	4
1.2	CHIPSET	5
1.3	HARDWARE BLOCK DIAGRAM	6
2	SYSTEM SPECIFICATIONS	6
2.1	GENERAL SPECIFICATIONS	6
2.2	WI-FI SPECIFICATION	6
2.3	SOFTWARE SPECIFICATION	7
3	REAR PANEL	8
4	BOTTOM PANEL	8
5	HOUSING	9
6	ACCESSORIES	9
7	POWER SUPPLY	9
8	ENVIRONMENT	9
9	CERTIFICATIONS	10
10	KEY COMPONENTS	10

1 Product Introduction

1.1 Overview

Internet Hub AX3000S is a WiFi6 ONT based on the IEEE802.11ax Wi-Fi technology. The Internet Hub AX3000S supports 2.4GHz 2ss and 5GHz 2ss dual-band, the total wireless link speed could reach up to 2975.5Mbps.



Figure 1: Internet Hub AX3000S Outlook (External antenna)

The WiFi6 ONT Internet Hub AX3000S supports the Wi-Fi Alliance standard EasyMesh, enabling the creation of a comprehensive home Wi-Fi network with multiple devices. The Internet Hub AX3000S function as a Controller, connecting to other Agents via a wired Ethernet cable or wireless through Wi-Fi. EasyMesh enhances the seamless roaming experience by allowing users to effortlessly switch between access points while maintaining a stable connection. This feature ensures uninterrupted connectivity as users move throughout the coverage area, resulting in a more reliable and user-friendly network experience.

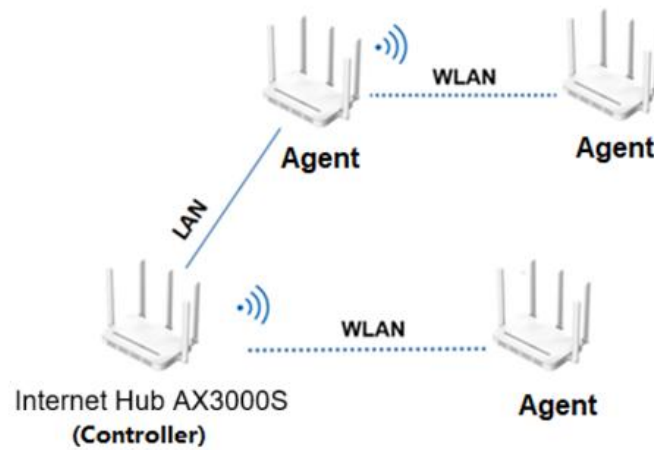


Figure 2: Internet Hub AX3000S Mesh Network Topology

1.2 Chipset

Name	Model
Main SoC	Airoha, AN7552CT, 1Ghz ARM CA53 Dual-Core
BoB chipset	EN7571N
FLASH	256MB SPI NAND
RAM	512MB DDR3
Wi-Fi baseband	MediaTek, MT7916A
Wi-Fi RF	MediaTek, MT7976DAN
FEM	2.4Ghz Wi-Fi: KCT8239SD 5Ghz Wi-Fi: KCT8539HE-1

1.3 Hardware Block Diagram

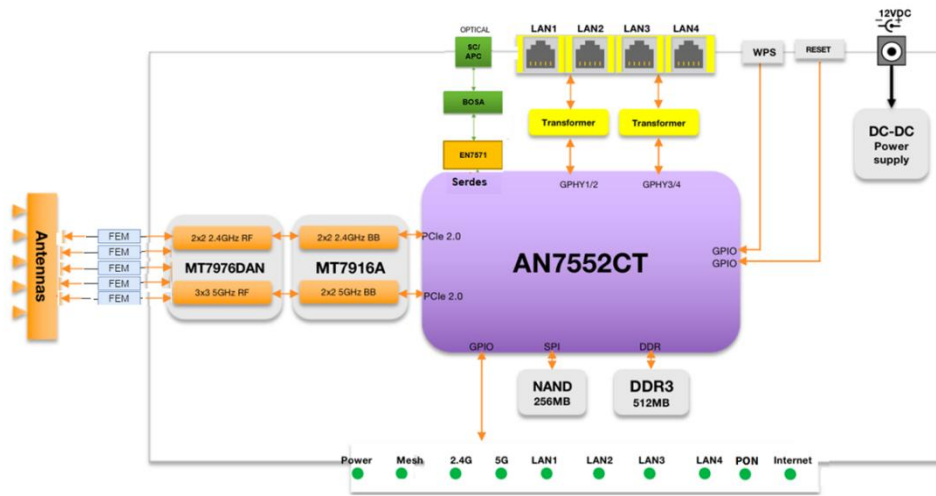


Figure 3: Internet Hub AX3000S Hardware Block Diagram

2 System Specifications

2.1 General Specifications

Interface	Description
Optical Interface	1*Optical Interface (SC/APC)
LAN Interface	4* Gigabit Ethernet RJ45 Ports
WLAN Interface	2x2 2.4Ghz base band, 2x2 5Ghz base band 2x2 2.4Ghz RF, 3x3 5Ghz RF

2.2 Wi-Fi Specification

Feature	Description
Compliant Standard	2.4Ghz: IEEE 802.11b/g/n/ax 5Ghz: IEEE 802.11a/n/ac/ax
Mesh	EasyMesh R4
Frequency band	2.4Ghz: 2400MHz-2483.5MHz (ch.1-13) 5Ghz: band1 (U-NII-1): 5150MHz-5250MHz (ch.36-48) band2 (U-NII-2A): 5250MHz-5350MHz (ch.52-64 DFS channel) band3 (U-NII-2C): 5490MHz-5730MHz (ch.100-144 DFS channel) band4 (U-NII-3): 5735MHz-5835MHz (ch.149-165)

MIMO	2x2 MU-MIMO for dual band
Channel BW	2.4Ghz: 20/40MHz 5Ghz: 20/40/80/160MHz
Data Rate (Maximum)	2.4Ghz - 573.5Mbps(11ax@40MHz CBW) 5Ghz - 2402Mbps(11ax@160MHz CBW)
Security	WPA/WPA-PSK & WPA2/WPA2-PSK, WPA3-SAE
Antenna	5*External Antennas, 5dBi
EIRP (Maximum)	2.4GHz: 22dBm 5GHz: 23dBm
Wi-Fi Channels	2.4G: 1~13
	5Ghz regular channels: 36, 40, 44, 48, and 149, 153, 157, 161, 165 5Ghz DFS channels.: 52, 56, 60, 64, 100, 104, 108, 112, 116, 120, 124, 128, 132, 136, 140, and 144

2.3 Software Specification

Feature	Description
Basic	PPPoE, DHCP, DNS, IGMP/MLD, NAT/NAPT, DNS-relay, SNTP
Advanced	Port Forwarding, ALG, UPnP, DDNS, SNTP, IGMPv2/v3 snooping&proxy, MLD v1/v2 snooping, ACL, VPN passthrough, NAT Loopback, QoS
Wi-Fi	OFDMA, MU-MIMO, 1024QAM, WPA3, BSS Coloring, Beamforming, EasyMesh, Band steering, AP steering, 802.11k/v/r
Firewall	ParentControl, MAC/IP/URL/Port Filtering, TCP SYN/ACK Flood/DoS Attack Prevention, DMZ Strange DDOS packet Filtering as: GRE, ESP, TCP/UDP
IP	IPv4 + IPv6 dual stack, DS-lite
Management	Local and Remote WEB management TR069, TR098, TR181(suggest), TR143 TR-369 User Services Platform (USP), MQTTv5.1 OTA upgrade via Web, and TR-069. System Logs

Security	Hash MD5 configuration file. Two level authentications for: Local & Remote Telnet/SSH/Console and Shell Strong password, WEB-GUI input validation.
----------	--

3 Rear Panel



Figure 4: Internet Hub AX3000S Rear Panel

4 Bottom Panel

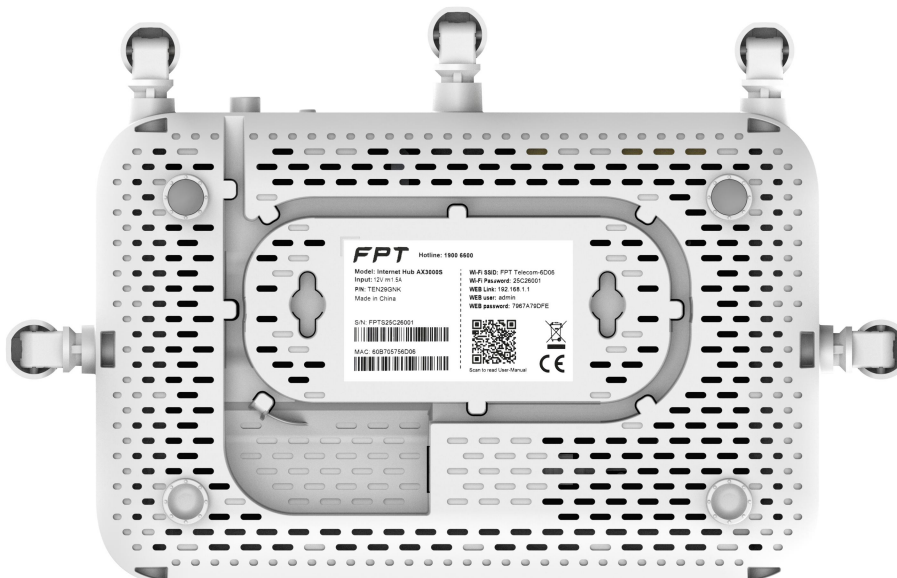


Figure 5: Internet Hub AX3000S Bottom Panel

5 Housing

Feature	Description
Housing Name	GNK
Housing Color	White
Dimension	205*134*38MM (W*H*D)
Net Weight	400g (device only)

6 Accessories

Feature	Description
Power Adapter	US Type, 1 *12VDC/1.5A
Network Cable	CAT5e, Grey, 1.5M

7 Power Supply

Feature	Description
Power Supply Mode	External Switching Mode Power Adaptor
AC Input (V/Hz)	AC 100~240V, 50~60Hz, 2-pin
Output Voltage(V)	12V DC 1.5A
Power Cable length	150cm
Power Consumption	<18W

8 Environment

Feature	Description
Operating Temperature	0℃ to 45℃ (32~113°F)
Storage Temperature	-20℃ to 65℃ (-4 ~149°F)
Operating Humidity	0 to 95% (non-condensing)
Operating Altitude	--60 to 2,134 meters

9 Certifications

Category	Standards Compliance
Safety	IEC/EN 62368-1 IEC/EN 62311 IEC/EN 60825-1 (Class 1 Laser Product)
EMC	EN 301 489-1/-17 EN 55032/EN 55035
WLAN Radio	EN 300 328 (2.4GHz) EN 301 893 (5GHz)
Certification	VNCE

10 Key Components

Component	Manufacturer	Part Number
SoC	Airoha	AN7552CT
BOB Chipset	Airoha	EN7571N
Flash	FORESEE	F35SQB002G
	MICRON	MT29F2G01ABAGDWB
RAM	ESMT	M15T4G16256A-DEBG2C
WiFi	MediaTek	MT7916A+MT7976DAN
FEM	KCT	2.4Ghz: KCT8239SD
		5Ghz: KCT8539HE-1
Power Adapter	AOHAI	YS-SKY120150U04P
	Ruijing	YS-SKY120150U04P